MOSAIC-Broncho: Multimodal Observation and State-Aware Imitation Control for Goal-Directed Autonomous Bronchoscopy

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***Abstract*— A bronchial bifurcation often looks clear to a human but remains ambiguous to a robot. The image alone does not reveal the intended branch, recent motion, or tendon hysteresis. We therefore present MOSAIC-Broncho, a Multimodal Observation and State-Aware Imitation Control framework for goal-directed bronchoscopy. A unified monocular front end extracts structural, anatomical, and metric-geometric cues. These cues are synchronized with motor position, finite-difference velocity, and frame timing to form an auditable 20-D state. The controller conditions an eight-frame history on the desired anatomical target and predicts bounded increments for insertion and two bending axes. It is trained from 87 real-robot demonstrations with 30,730 synchronized frames across seven targets. The resulting 773,221-parameter GRU reaches a held-out action MSE of 0.002084 at epoch 29. The study evaluates each link in the loop: perception quality, temporal and multimodal ablations, runtime behavior, and target-wise navigation in a silicone airway phantom. The design favors a small, inspectable state and continuous real-time control over a large end-to-end policy. It provides a practical route toward autonomous bronchoscopy when data, compute, and safety margins are limited.**

***Index Terms*— Medical robots and systems; imitation learning; recurrent neural networks; autonomous navigation; bronchoscopy; multimodal perception.**

I. Introduction

A bronchoscope reaches a fork. Both openings may look plausible. Yet only one leads to the target. The controller must remember the route, read the local geometry, and account for what the instrument has just done. A single image is not enough.

This compact example captures the central difficulty of autonomous bronchoscopy. Airway views are repetitive. Their appearance changes with reflection, mucus, deformation, and blur. The robot adds another source of ambiguity: tendon friction and hysteresis make the same command produce different motion histories [1], [2]. Although robotic platforms improve reach and stability, operation remains largely expert driven [3]–[5].

Recent work attacks different parts of this problem. Visual servoing keeps the lumen near the image center [6]. Online topological localization estimates where the scope is without patient-specific CT [7]. AI co-pilot learns shared control from a virtual expert and uses DAgger to reduce distribution shift [8]. BronchoCopilot fuses images and estimated pose in a reinforcement-learning agent [9]. More recently, long–short agents have combined a fast visual controller with slower strategic reasoning, reporting phantom, ex-vivo, and in-vivo evaluation [10]. Together, these systems show rapid progress. They also leave a practical gap. Simulation offers scale but creates transfer risk. Purely reactive control loses route context. Large or hierarchical policies can be difficult to run and audit at motor-loop rate.

Our premise is simple. Do not ask one image to carry the whole navigation problem. Instead, expose the few signals that matter and let a recurrent policy reason over their recent history. MOSAIC-Broncho follows this premise. Its MonoAirway-Fusion (MAF) front end turns a monocular frame into structural, anatomical, and metric-geometric evidence. The policy then combines these cues with the commanded goal and the measured motor state. It predicts continuous increments for insertion and two bending axes.

This is not a claim that a GRU is novel by itself. GRUs are established sequence models [11], and goal conditioning is well studied [12]. The contribution lies in the control formulation: a small, normalized, and auditable state connects monocular airway perception to real three-axis demonstrations. This design is deliberately data-efficient. It is also honest about the limits of behavior cloning, notably covariate shift outside expert-supported states [13]–[15].

We ask whether this compact formulation is sufficient for goal-directed navigation on a physical bronchial phantom. Our contributions are:

1) a unified monocular perception interface that exposes complementary structural, anatomical, and metric-geometric evidence;
2) a normalized 20-D state that joins this evidence with motor position, velocity, and timing, making every policy input loggable and inspectable;
3) a 773k-parameter, goal-conditioned GRU that maps eight-frame histories from real expert demonstrations to bounded three-axis commands; and
4) an evaluation design that connects component quality and controlled ablations to target-wise navigation on a physical silicone phantom.

II. Results

A. Unified Multimodal Perception

The structural output of MAF was evaluated for bronchial-junction segmentation in silicone airway images. It correctly delineated bifurcation regions in representative clean, low-light, specular, and blurred views. Quantitative reporting will use a fixed held-out test split and include Dice, IoU, precision, recall, and inference latency; these values are reserved for the recovered evaluation sheet rather than estimated from qualitative examples.

The nine-class anatomical output head contains 2.59M parameters. Its saved validation results are precision 0.8926,

TABLE I

MAF OUTPUT QUALITY IN SILICONE AIRWAYS

Output	Precision	Recall	Dice/mAP50	IoU/mAP50–95	Latency
Structure	[TODO: –]	[TODO: –]	[TODO: Dice]	[TODO: IoU]	[TODO: ms]
Anatomy	0.8926	0.7797	0.9096	0.6006	[TODO: ms]

TABLE II

MAF METRIC-GEOMETRIC EVALUATION

Output	N	MAE (mm)	RMSE (mm)	Bias (mm)
Raw	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
Calibrated	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]

recall 0.7797, mAP50 0.9096, and mAP50–95 0.6006. The gap between precision and recall motivates the temporal confirmation logic used by the runtime anatomical state estimator: isolated missed frames do not immediately invalidate the confirmed airway location.

B. Metric-Geometric Perception

The metric-geometric output of MAF was tested inside the laboratory silicone bronchial phantom at predefined camera-to-surface distances and at multiple airway locations. We report both raw model output and the linearly calibrated estimate. The final analysis will include sample count, mean absolute error (MAE), root mean square error (RMSE), signed bias, coefficient of determination, and valid-frame rate. Performance will also be stratified by normal illumination, specular reflection, partial occlusion, and motion blur.

C. Recurrent Policy Learning and Ablation

The real-robot dataset contains 87 expert demonstrations and 30,730 synchronized frames across seven recorded anatomical targets. The selected MOSAIC-GRU checkpoint contains 773,221 trainable parameters. It was selected at epoch 29 with a held-out action MSE of 0.0020837828 and auxiliary task accuracy of 1.000. The first logged validation action MSE was 0.02577, corresponding to a 91.9% reduction at the selected checkpoint. Task accuracy must not be interpreted as navigation success because it measures only the recorded navigate/intervene label.

The ablation study evaluates the full model against a frame-wise MLP, a GRU without goal conditioning, a single-head GRU, a model without depth/path features, and a model without velocity/timing features. Every variant uses the same trajectory-wise split and reports total and per-axis action error, macro-F1, parameter count, and hardware route success.

D. Real-Robot Navigation Success

The complete policy was evaluated from a fixed start pose in the silicone bronchial phantom. A trial is successful only if the bronchoscope enters the predefined target branch without operator takeover and remains there for the confirmation interval. We report per-target and pooled success with Wilson 95% confidence intervals, together with wrong-branch rate, completion time, contact count, and intervention count.

TABLE III

MOSAIC-GRU ABLATION

Variant	Action MSE	Task F1	Parameters	Route success
Frame-wise MLP	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
GRU, no goal	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
GRU, action only	[TODO: –]	–	[TODO: –]	[TODO: –]
No depth/path	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
No velocity/Δt	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
Full model	0.002084	[TODO: macro-F1]	773,221	[TODO: –]

TABLE IV

REAL-ROBOT NAVIGATION RESULTS

Target	Success (n/N)	95% CI
LMB	[TODO: –/–]	[TODO: –]
RMB	[TODO: –/–]	[TODO: –]
LUB	[TODO: –/–]	[TODO: –]
LLB	[TODO: –/–]	[TODO: –]
RUL	[TODO: –/–]	[TODO: –]
RML	[TODO: –/–]	[TODO: –]
RLL	[TODO: –/–]	[TODO: –]
Pooled	[TODO: –/–]	[TODO: –]

Together, these four result groups evaluate the complete perception-to-action chain: unified multimodal perception, recurrent policy learning, controlled ablation, and physical target arrival. Values shown as placeholders require the corresponding saved evaluation sheet and are not inferred from videos.

III. SYSTEM AND MULTIMODAL OBSERVATION

A. Robotic Platform

The platform uses a flexible bronchoscope driven by three position-controlled motors: insertion/withdrawal M_0, left–right bending M_1, and up–down bending M_2. The software separates camera/perception and motor-control threads through a thread-safe feature dictionary. Commands are checked against the implemented limits $M_0 \in [-900, 0]^\circ$, $M_1 \in [-170, 170]^\circ$, and $M_2 \in [-600, 600]^\circ$. The policy interface accepts the newest perception snapshot and cached motor feedback and can be called at 20–60 Hz.

B. MonoAirway-Fusion Perception

MAF treats monocular perception as one structured inference operator. For an RGB frame I_t, it exposes structural evidence $\mathbf{s}_t$, anatomical evidence $\mathbf{c}_t$, and metric-geometric evidence $\mathbf{d}_t$:

$$(\mathbf{s}_t, \mathbf{c}_t, \mathbf{d}_t) = \Phi_{\mathrm{MAF}}(I_t). \tag{1}$$

The structural stream provides obstruction and bifurcation centroids, areas, and validity; the anatomical stream provides airway class and confidence; the metric-geometric stream provides center, mean, and maximum depth together with left, center, and right free-space summaries. The streams

TABLE V

STRUCTURED OBSERVATION USED BY THE DEPLOYED POLICY

Group	Dim.	Contents
Target/junction	7	centroids, areas, detection flag
Depth geometry	6	center/mean/max, left/center/right
Motor position	3	M_0, M_1, M_2 angles
Motor velocity	3	finite-difference angular velocity
Timing	1	frame interval Δt
Total	20	normalized structured state

form a single 13-dimensional visual-geometric interface rather than high-dimensional maps.

For reproducibility, the present implementation instantiates the structural stream with a VGG–U-Net segmentation backbone [16], the anatomical stream with a lightweight YOLO-family detector [17], and the monocular-depth prior with Depth Anything V2 [18]. These are replaceable estimators inside MAF; the method-level contribution is their synchronized, physically interpretable interface and its coupling to recurrent control.

Let $\mathbf{v}_t = [\mathbf{s}_t; \mathbf{c}_t; \mathbf{d}_t] \in \mathbb{R}^{13}$. For each time t, $\mathbf{o}_t \in \mathbb{R}^{20}$ contains seven visual features, six depth/path features, three motor angles, three finite-difference motor velocities, and the elapsed time Δt. Each component is normalized by a fixed physical or geometric scale r_i:

$$\tilde{o}_{t,i} = o_{t,i}/r_i, \qquad \tilde{\mathbf{o}}_t = [\mathbf{v}_t; \mathbf{q}_t; \dot{\mathbf{q}}_t; \Delta t]. \tag{2}$$

The resulting representation is small enough to inspect, log, and replay, while temporal ambiguity is delegated to the recurrent policy.

IV. MOSAIC GOAL-CONDITIONED RECURRENT POLICY

A. Architecture

The policy receives an eight-frame window $\mathbf{O}_t = [\mathbf{o}_{t-7}, \ldots, \mathbf{o}_t]$ and a goal $g \in \{0, \ldots, 9\}$. Goals comprise exploration, trachea, and eight named bronchial destinations. A shared multilayer perceptron encodes each observation through $20 \to 128 \to 128$ layers with LayerNorm and ELU. A learned lookup table maps the goal to a 16-dimensional vector, which is replicated across time and concatenated with every encoded observation.

The 144-dimensional sequence is processed by a two-layer GRU with hidden size 256 and inter-layer dropout 0.1:

$$\mathbf{x}_t = [f_{\text{obs}}(\tilde{\mathbf{o}}_t); f_{\text{goal}}(g)], \tag{3}$$

$$\mathbf{h}_t = \text{GRU}(\mathbf{x}_t, \mathbf{h}_{t-1}). \tag{4}$$

The final state feeds an action head $256 \to 128 \to 3$ with $\tanh$ output and a task head $256 \to 64 \to 2$. The action head predicts normalized increments for insertion and two bending axes. The task head distinguishes navigation from obstruction intervention and acts as an auxiliary representation constraint.

TABLE VI

DESIGN CONTEXT IN AUTONOMOUS BRONCHOSCOPY

Method	Training	Online input	Memory	Validation
AI co-pilot [8]	sim. imitation	RGB + command	DAgger	phantom/in vivo
BronchoCopilot [9]	sim. RL	RGB + pose	latent state	simulation
Long–short [10]	hierarchical	RGB + CT goal	dual agent	phantom/ex vivo/in vivo
MOSAIC-Broncho	real BC	20-D + goal	8-frame GRU	silicone phantom

B. Learning Objective and Data Split

The action and auxiliary objectives are

$$\mathcal{L} = \|\hat{\mathbf{a}}_t - \mathbf{a}_t^E\|_2^2 + \lambda_s \, \text{CE}(\hat{\mathbf{z}}_t, s_t^E), \tag{5}$$

where $\mathbf{a}_t^E$ is the expert motor increment and s_t^E is the task label. Training uses AdamW, OneCycleLR, mixed precision, gradient clipping, and batch size 64. Demonstrations are split by whole trajectory (90/10), preventing adjacent windows from the same run from entering both training and validation.

The repository contains 87 real-robot demonstrations and 30,730 synchronized frames across seven recorded anatomical targets. Windows are sampled with $T = 8$. Actions remain normalized during learning; deployment scales $\tanh$ outputs by $[5, 3, 3]^\circ$ per frame for $M_0/M_1/M_2$ and applies an additional configurable safety factor. The normalized action is converted to a safe absolute target as

$$\mathbf{q}_{t+1}^{\text{cmd}} = \text{clip}(\mathbf{q}_t + \beta \, \boldsymbol{\alpha} \odot \hat{\mathbf{a}}_t, \mathbf{q}_{\min}, \mathbf{q}_{\max}), \tag{6}$$

where $\boldsymbol{\alpha} = [5, 3, 3]^\circ$ and $\beta \in (0, 1]$ is the deployment safety factor.

C. Stateful Real-Time Inference

The runner maintains a fixed-length deque, the goal token, and recurrent hidden state. At each control step it (i) obtains the newest synchronized feature dictionary, (ii) estimates velocities from cached angles and Δt, (iii) normalizes the 20-dimensional observation, (iv) performs inference without gradients, and (v) converts bounded increments to absolute motor targets. When the goal changes or autonomy restarts, the buffer and hidden state are reset. Missing fields fall back to neutral values, while the surrounding controller uses timestamps and consecutive-frame confirmation to prevent stale visual events from triggering mode changes.

Table VI places the method among its closest autonomous bronchoscopy counterparts. It is a design comparison, not a performance ranking, because the systems use different robots, targets, and test domains.

V. EXPERIMENTAL PROTOCOL

A. Phantom Evaluation

The final study will use paired, randomized trials from a fixed start jig on the same airway phantom. Targets include LMB, RMB, LUB, LLB, RUL, RML, and RLL. Initial pose, illumination, lens contamination, and mucus placement are varied according to a predefined schedule. The primary comparison is the proposed goal-conditioned GRU against (B0) expert teleoperation, (B1) frame-wise behavior cloning, (B2) GRU without goal conditioning, and (B3) reactive visual servoing.

Primary outcomes are route success and wrong-branch rate. Secondary outcomes are normalized image-plane center error, wall-contact count and duration, completion time, action jerk, and inference/control latency. Each target and condition will report trial count, Wilson 95% confidence intervals for binary outcomes, median and interquartile range for skewed continuous metrics, and a paired test with correction for multiple comparisons. Failures are classified as wrong branch, timeout, perception loss, contact stop, or operator takeover.

B. Ablations

Four ablations isolate the source of performance: remove the goal embedding; replace the GRU with a frame-wise MLP; remove depth/path features; and remove motor velocity and Δt. A fifth experiment compares the dual-head loss against action-only behavior cloning. These ablations test whether improvements arise from goal conditioning, temporal memory, geometric perception, dynamic state, or auxiliary supervision rather than parameter count alone.

VI. Discussion

MOSAIC-Broncho occupies a middle ground between reactive servoing and large hierarchical agents. It retains explicit geometry and robot state, but learns how their recent history maps to action. This choice reduces model size and makes failures easier to trace. It also limits semantic breadth. The policy can only pursue configured anatomical goals, and its competence is bounded by the expert trajectories it has seen.

The comparison with recent systems is therefore nuanced. AI co-pilot has stronger sim-to-real training and in-vivo evidence [8]. BronchoCopilot demonstrates deeper simulated navigation with multimodal RL [9]. Long–short agents address long-horizon planning and report ex-vivo and in-vivo experiments [10]. Our current strength is different: the policy learns continuous three-axis control from real robot demonstrations and operates on a compact, inspectable temporal state. This is useful, but it is not yet evidence of state-of-the-art navigation performance.

The present evidence has limitations. Validation MSE measures agreement with recorded actions, not recovery from covariate shift. Seven of ten configured goals have demonstrations, class balance is uneven, and perfect auxiliary accuracy may reflect label simplicity. Monocular depth can drift under fluid, specularity, and deformation. Finally, physical route success, contacts, and latency distributions remain to be inserted from randomized trial logs before submission. These limitations are made explicit to prevent software tests or qualitative videos from being misreported as efficacy evidence.

VII. Conclusion

We presented MOSAIC-Broncho, a compact imitation-control framework for goal-directed robotic bronchoscopy. Its key idea is straightforward: make the navigation state explicit, then reason over time. A 20-D observation captures monocular airway cues and robot dynamics. An eight-frame, goal-conditioned GRU turns that history into bounded three-axis commands. The current artifacts establish the data scale, model design, and held-out prediction error. The locked phantom protocol defines the remaining evidence needed for submission. If the physical trials confirm robust target arrival, the system will show that useful bronchoscopic autonomy does not always require a large end-to-end model.

Acknowledgment

For double-anonymous review, institutional support is omitted. OpenAI Codex was used to assist with manuscript organization, language drafting, and LaTeX preparation. All technical claims, citations, equations, and reported values were checked by the authors against the project code and saved artifacts.

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